

LEWISSETTA, VA, USA

WMO: 997304

Lat: 37.9954N

Lon: 76.4646W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	23.3	N/A	N/A	N/A	N/A	N/A	N/A	28.5	40.0	26.6	37.0	11.8	350	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	90.2	N/A	88.0	N/A	86.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.3	230

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
26.9	23.2	19.8	DB	15.9	94.7	4.8	2.1	12.5	96.2	9.7	97.4	7.0	98.5	3.5	100.0	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	38.1	39.9	46.5	58.1	66.9	75.5	79.5	77.9	72.1	62.1	50.4	43.7
	DBStd	16.23	9.64	9.02	8.82	7.83	6.69	5.08	3.76	3.56	5.04	7.11	7.33	8.46
	HDD50	1207	387	305	177	16	0	0	0	0	0	5	82	235
	HDD65	3649	833	703	576	236	59	1	0	0	5	138	437	661
	CDD50	4612	19	22	68	258	525	764	913	865	663	380	95	40
	CDD65	1581	0	1	2	28	119	315	448	400	218	48	1	1
	CDH74	11427	0	0	9	159	579	2205	4102	3169	1063	141	0	0
	CDH80	3026	0	0	0	31	112	583	1301	790	196	13	0	0
Wind	WSAvg	9.4	9.8	10.0	10.4	10.3	9.5	8.6	8.6	8.2	8.8	9.6	9.5	9.1
Precipitation	PrecAvg	46.3	3.3	2.9	4.0	3.3	3.7	4.0	5.2	4.6	3.7	2.8	3.5	
	PrecMax	65.9	6.9	7.8	9.1	6.1	7.1	8.1	9.1	11.7	16.1	7.5	9.0	8.1
	PrecMin	35.0	1.3	0.9	0.3	1.4	0.9	1.7	1.4	1.7	0.6	0.1	0.2	1.3
	PrecStd	7.0	1.3	1.5	1.6	1.1	1.5	1.4	1.9	2.6	3.4	1.9	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	69.7	75.2	84.2	87.2	92.3	94.5	91.4	88.7	81.3	71.7	70.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	63.0	64.3	69.3	78.4	83.3	87.9	91.2	88.7	85.0	77.4	67.7	66.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	58.7	59.1	64.9	74.1	79.9	85.8	88.7	86.5	82.1	75.1	64.4	61.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	53.4	53.1	60.2	70.2	77.2	83.4	86.5	84.3	79.7	72.7	61.7	56.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.5	12.8	13.3	12.1	11.1	11.1	10.0	9.7	10.9	11.2	11.1
		MCDBR	19.5	20.3	21.1	18.9	15.2	14.3	14.3	13.2	12.8	13.0	14.9	16.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.309	0.319	0.352	0.392	0.433	0.483	0.515	0.508	0.410	0.358	0.325	0.312
	taud		2.527	2.490	2.438	2.339	2.262	2.166	2.092	2.097	2.351	2.496	2.545	2.554
	Ebn at Noon		279	290	289	281	271	256	246	245	266	272	270	270
	Edh at Noon		24	29	33	39	43	47	51	49	36	28	24	23
All-Sky Solar Radiation	RadAvg		710	962	1286	1623	1846	2018	1946	1748	1414	1089	803	626
	RadStd		55	103	106	116	193	147	94	93	137	121	69	53

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	+169

Nomenclature: See separate page